

This thesis was submitted to the Chair of Machine Learning and Reasoning.

Title of Your Thesis

Master Thesis

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Aachen, February 26, 2026

Abstract

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0 Template Instructions

Read the README before you start here. Then take a look at the below examples, both in the PDF and in the source code, to get an idea of how to use this template. Delete this chapter before your final submission.

0.1 Chapters and Sections

You can begin new chapters and sections with the `\chapter` and `\section` commands. This goes down over `\subsection` to the `\subsubsection` level.

Do not skip any of the section levels and always include at least one sentence between successive headings. Also, use title case consistently for all headings. If you are unsure, consult the Title Case Converter at <https://titlecaseconverter.com>.

0.2 Acronyms

Acronyms can be included with the `\ac` command and are fully written out on first use, for example Rheinisch-Westfälische Technische Hochschule (RWTH). On subsequent uses, only the short version is inserted, like RWTH.

For plural versions, use `\acp`, giving RWTHs. If you want to force the long or short version anywhere, use `\acl` or `\acs`. Don't use acronym commands in chapter or section titles.

Do not use the acronym commands inside `\chapter` or `\section` commands! This breaks compilation when the headings are cross-referenced with `\cref`. Just write the acronym or long version out in plain text.

0.3 Citations

Use the `\cite` command to insert citations, like [1]. Multiple citations can be added with comma separation in a single command. It is recommended to include a non-breaking space before the citation with `~`.

To insert the authors Sutton and Barto or year 2018 use `\citeauthor` and `\citeyear`. With `\textcite` you can insert a combined author and number citation like Sutton and Barto [1].

0.4 Cross-References

Insert cross-references to other sections, figures, equations, etc. with the `\cref` command. This includes not only the number but also the text part, for example Section 0.1, Figure 2, Table 1, Equation (1), Listing 2 or Algorithm 1.

Multiple references can be added with comma separation (without spaces) in a single `\cref` command, like Figures 1 and 2.

0.5 SI Units

Use `\num` and `\qty` commands for pretty printing of numbers and units. For example, large numbers like 42 000, percentages like 20 %, or ranges like 1.5 m s^{-1} to 2.5 m s^{-1} and lists like 1.3 mm, 6.7 mm, and 8.0 mm.

0.6 TODO Notes

This is a remark

You can add remarks in the page margin with the `\TODO` command. Inline remarks are also possible, see below, or you can hide the line.

This is an inline remark

Another remark

An overview of all remarks is printed in the list of TODOs at the end of the document. The remarks and the list disappear if you add the `final` option to the `\documentclass` command in the preamble.

0.7 Equations

This template is set up to replace `(, )`, `[, and ]` in math mode with `\left(, \right)`, `\left[, and \right]`. This means you don't have to worry about parentheses and bracket size adjustments. Just use them naturally like

$$(1 - a) \cdot \left(1 - \frac{b}{c}\right). \quad (1)$$

For bold symbols, use the `\symbf` command that works for both Latin and Greek letters like **x** and **ϕ** (as opposed to `\mathbf`). It also fixes some alignment issues, e.g. **Ã** versus **Ã**. Similarly, you can use `\symcal` or `\symbb` and other commands instead of their `\math...` counterparts.

Keep in mind that equations are part of the sentence, so punctuation like full stops and commas still need to be inserted after equations like this geometric series

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q} \quad \text{with } |q| < 1. \quad (2)$$

For proofs, use the `\begin{proof} ... \end{proof}` environment. This can be combined with the `\begin{theorem} ... \end{theorem}` environment if applicable.

Theorem 1 (Even Squares). *For all integers n , if n is even, then the square of n is even.*

Proof. Let n be any even integer. Then $n = 2k$ for some integer k and

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, n^2 is even. □

0.8 Listings

Source code listings can be included from files with the `\lstinputlisting` command or directly inside `lstlisting` environments. You can specify captions and labels as well. Additionally specify the language for basic syntax highlighting.

Listing 1: Caption.

```
1 #include <iostream>
2
3 int main() {
4     std::cout << "Hello World!";
5     return 0;
6 }
```

Listing 2: Caption.

```
1 print("Hello World!") # comment
```

0.9 Floating Elements

Figures, tables and algorithms are floating elements in $\text{T}_{\text{E}}\text{X}$. Here are some examples, mostly just to be referenced from Section 0.4.

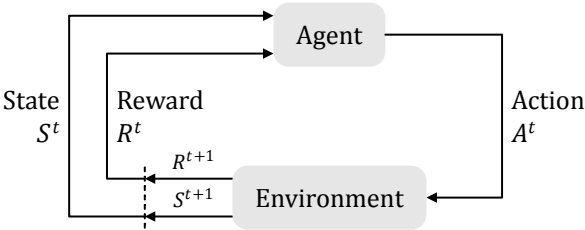


Figure 1: The agent-environment interaction in a Markov decision process (MDP), adapted from Sutton and Barto [1]. In each time step, the agent observes a state S^t and a reward R^t and chooses an action A^t . This leads to a change in the environment and a new state S^{t+1} and reward R^{t+1} are issued.

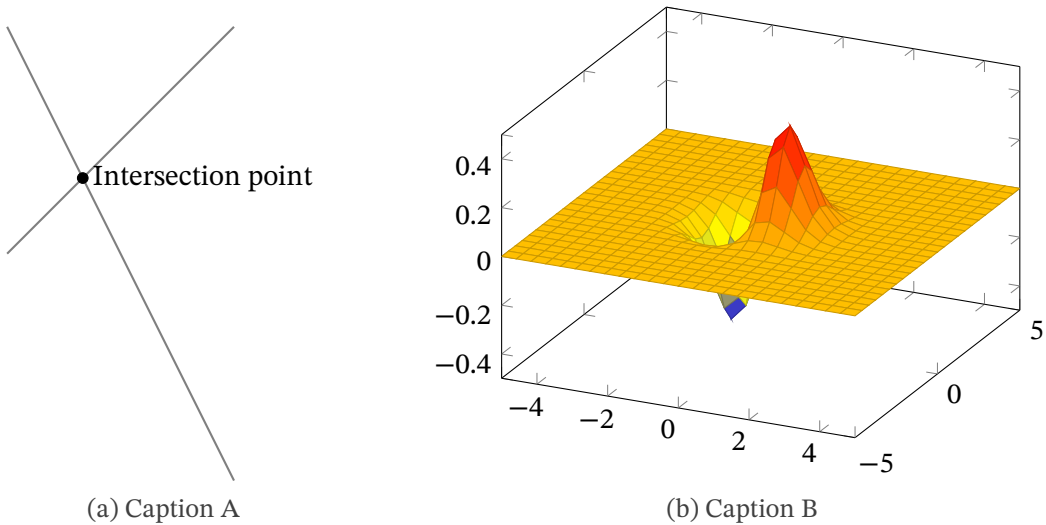


Figure 2: Caption.

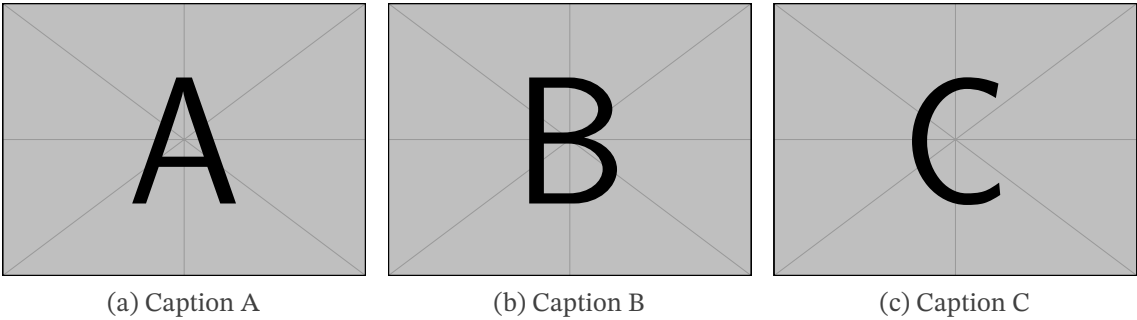


Figure 3: Caption.

Table 1: Caption.

Column A	Column B	Column C
Row 1	Text	42.0
Row 2	Text	31.4

Table 2: Caption.

Column A	Column B	Column C
Row 1	Text	42.0
Row 2	Text	31.4

Algorithm 1 Greatest common denominator.

1	procedure EUCLID(a, b)	<i>Finds the g.c.d. of a and b</i>
2	$r \leftarrow a \bmod b$	
3	while $r \neq 0$ do	<i>We have a result if $r = 0$</i>
4	$a \leftarrow b$	
5	$b \leftarrow r$	
6	$r \leftarrow a \bmod b$	
7	return b	<i>The g.c.d. is b</i>

1 Introduction

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2 Background

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3 Related Work

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4 Methods

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5 Results

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6 Conclusion

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A Appendix

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List of Acronyms

MDP	Markov Decision Process
PDF	Probability Density Function
RV	Random Variable
RWTH	Rheinisch-Westfälische Technische Hochschule

List of Symbols

General

$(\cdot)$	arbitrary (scalar) variable
$(\bullet)$	arbitrary vector variable
$\approx$	approximately equal
$\langle \cdot, \cdot, \cdot \rangle$	tuple of multiple variables
$\mathbb{R}$	set of real numbers
$\mathbb{E}[X]$	expected value of a random variable (RV) X
$\mathbb{P}[X = x]$	probability that an RV X takes on the value x
$\frac{\partial f(\cdot)}{\partial x}$	partial derivative of f with respect to x
$\left. \frac{\partial f(\cdot)}{\partial x} \right _a$	partial derivative of f with respect to x evaluated at a
$\arg \min_a f(a)$	optimal value of a minimizing the expression $f(a)$
$\arg \max_a f(a)$	optimal value of a maximizing the expression $f(a)$
π	mathematical constant, $\pi \approx 3.142$
$\mathcal{N}(\mu, \sigma)$	Gaussian normal distribution with mean μ and standard deviation σ

Markov Decision Process

t	time step index
$\mathcal{S}$	set of all states
$\mathcal{A}$	set of all actions
$\mathcal{R}$	set of all possible rewards
s, s^t	state at time t , realization of RV S^t
s', s^{t+1}	next state at time $t + 1$, realization of RV S^{t+1}
a, a^t	action at time t , realization of RV A^t
a', a^{t+1}	next action at time $t + 1$, realization of RV A^{t+1}
r, r^{t+1}	reward at time $t + 1$, realization of RV R^{t+1}
S^t	state at time t
A^t	action at time t
R^t	reward at time t
$p(s', r \mid s, a)$	dynamics function specifying a conditional probability density function (PDF) for the next state s' and reward r given the current state s and action a

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2	Caption.	3

List of References

- [1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, second edition. MIT Press, Nov. 13, 2018, ISBN: 978-0-262-03924-6.

List of TODOs

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This is an inline remark	2
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